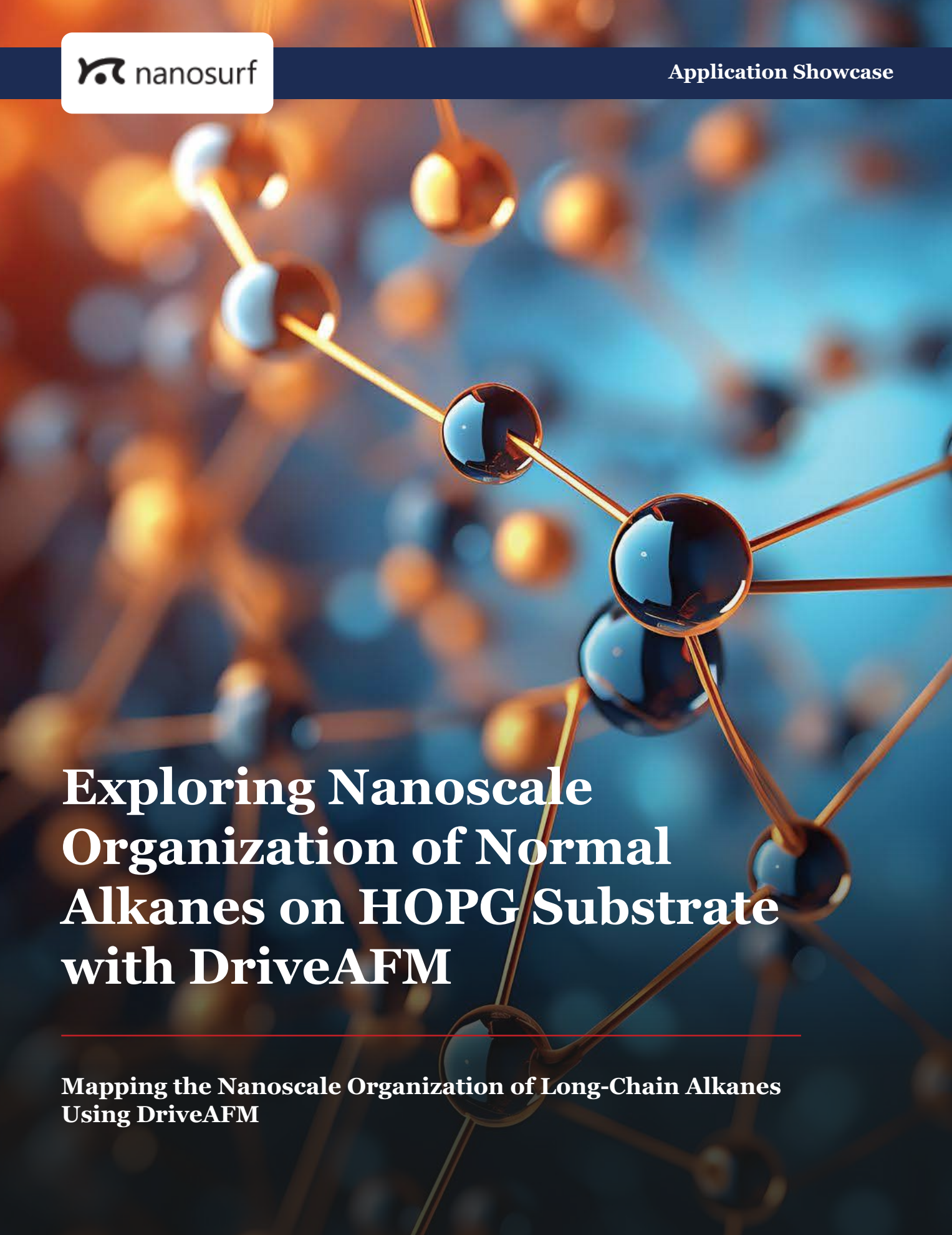


A detailed 3D molecular model of a long-chain alkane molecule, likely n-octane, is shown against a blurred background of other similar molecules. The atoms are represented by spheres (carbon in dark blue, hydrogen in light blue) connected by thin rods. The lighting creates a warm, golden glow, highlighting the molecular structure.

Exploring Nanoscale Organization of Normal Alkanes on HOPG Substrate with DriveAFM

Mapping the Nanoscale Organization of Long-Chain Alkanes Using DriveAFM

INTRODUCTION

Normal alkanes C_nH_{2n+2} are linear chain molecules, which are typically extended, and they form ordered lamellar structures and crystals. When size of the chains increases with n above 120, such ultra-long alkanes manifest chain folding – the main driver behind of polymer crystallization. The longest alkane $C_{390}H_{782}$ is a perfect model of linear polyethylene, and crystallization of these species have many common features.

The lamellar 2D organization has been detected in thin and ultra-thin layers of alkanes on highly oriented pyrolytic graphite (HOPG) and other layered crystals such as MoS₂. On these substrates the molecular orientation exhibits three-fold symmetry that guides the related orientation of the alkane lamellae. As the lamellae are formed of fully elongated chains, their width equals to the alkane length when the molecules are aligned perpendicular to the lamellar edges. The length of different alkanes from C₁₈H₃₈ to the ultra-long one – C₃₉₀H₇₈₂ changes as listed in Figure 1a. The lamellar width can be smaller when molecules are tilted at some angle to the lamellar edges. Figure 1b shows a sketch of the alkane lamellae that should have thickness of individual alkane molecule (0.5 nm).

The lamellar order of all these alkanes on HOPG can be revealed in AFM images at room temperature as melting temperature of C₁₈H₃₈ crystals is around 28–30°C and longer alkanes melt at higher temperatures. It is known that lamellar ordering of alkanes, which is facilitated by inter

The nanoscale self-organization of normal and ultra-long alkanes (C_nH_{2n+2}) on highly oriented pyrolytic graphite (HOPG) was investigated using the DriveAFM microscope equipped with photothermal excitation (CleanDrive) and WaveMode imaging. High-resolution topography and phase data revealed distinct lamellar architectures whose periodicities correspond closely to the molecular chain lengths of the alkanes studied (C₆₀H₁₂₂, C₁₂₂H₂₄₆, C₂₄₂H₄₈₆, and C₃₉₀H₇₈₂). Subtle topographic modulations of ~0.2 nm were attributed to alternating –CH₃ and –CH₂– groups at lamellar edges, while variations in tip-sample interaction strength exposed hidden periodicities arising from lateral chain folding. These findings demonstrate the sensitivity of DriveAFM in capturing fine variations in lamellar order, molecular orientation, and chain packing that are otherwise inaccessible with conventional imaging. The results highlight how AFM can elucidate the structural dynamics underlying alkane crystallization and provide nanoscale insights relevant to polymer morphology and self-assembly phenomena.

Keywords or phrases:

Atomic Force Microscopy (AFM), Normal Alkanes, Lamellar Structures, HOPG Substrate, Molecular Self-Organization, Chain Folding

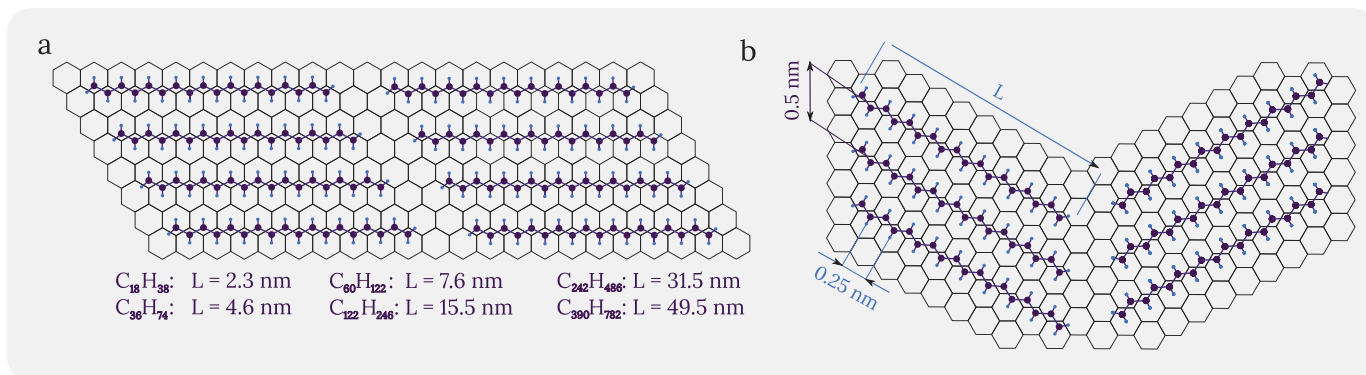


Figure 1.: Self-organization of normal alkane molecules (C_nH_{2n+2}) on HOPG surface. The HOPG surface is depicted as a honeycomb lattice, the carbon atoms are dark purple circles and the hydrogen atoms are light blue circles. (a) Horizontal packing with molecular chains in different lamellae parallel to each other. (b) V-shaped packing, with the molecules in different lamellae positioned at angle to each other. Text below the sketch (a) indicates lengths L of alkane molecules of different molecular weights.

molecular interactions on atomically flat surfaces like HOPG, is preserved at temperatures higher than melting temperature of alkane crystals. This was confirmed by detection of $C_{390}H_{782}$ lamellar layers on HOPG at temperatures 50 degrees above melting temperature of these alkanes (130°C).

The lamellar edges are formed of $-CH_3$ end groups, which are bulkier than $-CH_2-$ groups of the chains. The related height variations should be reflected in topography of lamellar sheets formed on atomically flat substrates. This application demonstrates how the lamellar structures of normal alkanes with different chain length on HOPG are visualized with AFM and what peculiar nanoscale features can be detected.

EXPERIMENTAL DETAILS

The samples of different alkanes were prepared by spin-casting of their solutions in toluene (concentration is around 10 mg/ml) on freshly cleaved surface of HOPG. All AFM measurements were performed at room temperature and in most cases, we have used DriveAFM microscope in Dynamic Mode by using photo-thermal excitation of the probe (CleanDrive) instead of piezo-electric drive. Few studies were also made in WaveMode, in which intermittent contact has proceed at off-resonance frequency of 3 kHz. Commercially available soft dynamic mode cantilevers with spring constants in the 1-10 N/m range were applied in these experiments.

Imaging of normal alkanes: $C_{60}H_{122}$

Visualization of lamellar order of normal alkane $C_{60}H_{122}$ with molecular length of 7.5 nm is displayed in Figures 2a-d. An area of 1 μm on side, which is shown in Figure 2a, consists of several terraces covered by barely seen fine linear features. Several nanoscale size droplets most likely represent an alkane material, which was not spread on the substrate. The height image recorded at a smaller location in the center of this area reveals a periodical pattern with the spacing of 7.5 nm. Height and phase images in Figure 2c-d present a location with two HOPG crystalline terraces, which displays

the periodical arrays of linear features with different orientations. The width of these features corresponds to $C_{60}H_{122}$ length that allows their assignment to the alkane lamellae. The height depressions, which are noticed perpendicular to the lamellar edges on the lower terrace, are ~ 0.2 nm in height. These features can be assigned to $-CH_3$ terminals that are slightly higher than $-CH_2-$ groups of the lamellar sheet. However, the end $-CH_3$ groups are more mobile than $-CH_2-$ groups of the chain, and they can be stronger depressed by the probe tip. These local stiffness variations are most likely responsible for the phase contrast in Figure 2d.

Imaging of normal alkanes: $C_{122}H_{246}$

The multiple lamellar sheets of $C_{122}H_{246}$ alkanes cover the surfaces of two substrate terraces, which are separated by a diagonal border in the area displayed in the height image, Figure 3a. The wavy structures of 15-16 nm in width are distinguished in the area on right from the border, Figure 3b. They represent $C_{122}H_{246}$ lamellae as their width is close to $C_{122}H_{246}$ molecular length. The strips with similar width have formed the raised ribbon with height of 3.5 nm.

Several overlaying strips have composed the ribbon, and they oriented like the wavy structures occupied most of the area. Several dark blocks $\sim 15-16$ nm in width represent the missing parts of the lamellae. The terraces of multiple overlaying alkane lamellae, which are noticed on the left from the border, are displayed in the height image in Figure 3c. The steps between top layers are ~ 0.5 nm that consistent with diameter of individual alkane molecule. Faint contrast variations in this image hints on a regular surface order with the periodical spacing close to the alkane width. To improve structure visualization, we have increased the tip-sample force interactions by lowering set-point amplitude.

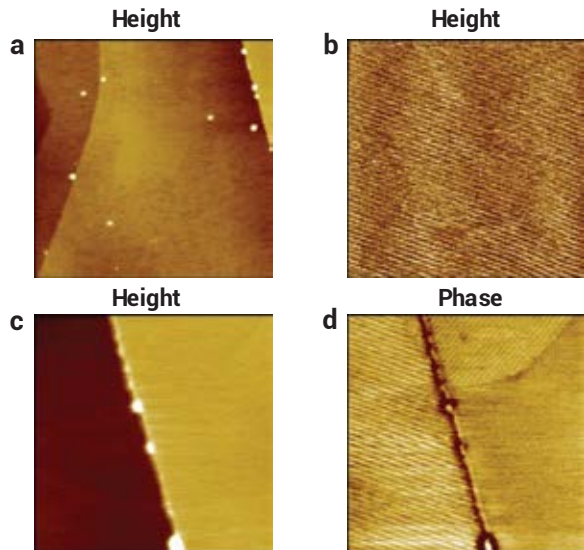


Figure 2.: Height (a,b,c) and phase (d) images of C₆₀H₁₂₂ layer on HOPG. (a) Terraces of HOPG with fine line features representing the alkane assemblies, small droplets are likely the alkane material, not spread on the surface. (b) Zoom in the center of the image (a) showing a periodic pattern of alkane assemblies with the period of 7.5 nm. (c,d) Height and phase images showing two HOPG crystalline terraces with the arrays of linear features with different orientations.

(a) Image size: 1 x 1 μm^2 , height range: 2.4 nm.
 (b) Image size: 0.4 x 0.4 μm^2 , height range: 1 Å.
 (c,d) Image size: 0.4 x 0.4 μm^2 , height range: 1.8 nm, phase range 48°.

The related effect is obvious in height image in Figure 3d, which was recorded in the scanning from the top. At the top, scanning was performed at small tip force. Then the tip-force was raised, and the height contrast was drastically improved. The linear features became resolved not only with the spacing of 15–16 nm but also at smaller periodicities. To verify the observation was verified in the bottom of the image when the set-point was raised again and lowered. The related image changes have supported the effect.

The image pattern revealed at high-force imaging can be explained by double folding of alkane molecules that leads to appearance additional spacing of 5 nm. This situation is sketched in Figure 3f.

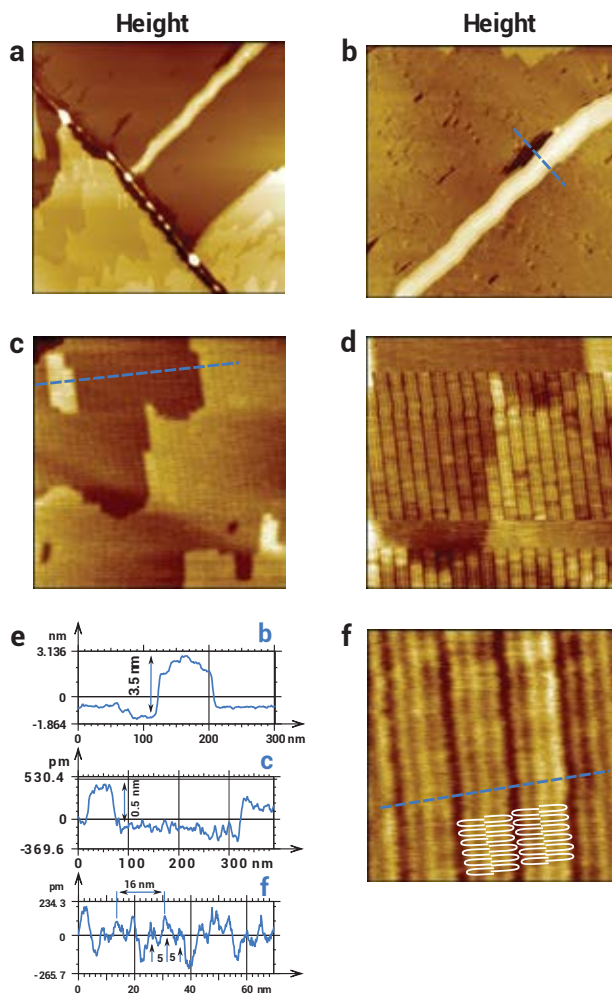


Figure 3.: Height images of C₁₂₂H₂₄₆ layer on HOPG. The low tip-force interactions were used for images in (a)–(c). The image in (d) was recorded at low and high tip-forces: the scanning started at the top of the image at low force, later the force was increased, and the visibility of lamellar structures has improved. (e) The cross-sections along the blue dashed lines in images (b,c,f). (f) A tentative pattern of lateral chain folding was overlaid on the height image.

(a) Image size: 1.5 x 1.5 μm^2 , height range: 7 nm.
 (b) Image size: 1 x 1 μm^2 , height range: 5 nm.
 (c) Image size: 0.5 x 0.5 μm^2 , height range: 3 nm.
 (d) Image size: 0.3 x 0.3 μm^2 , height range: 1.2 nm.
 (f) Image size: 76 x 76 nm², height range: 0.5 nm.

Imaging of normal alkanes: C₂₄₂H₄₈₆

Folding of macromolecules is common for single crystals of ultra-long alkanes and folding of C₃₉₀H₇₈₂ chain in 5 segments leads to 10 nm thick lozenge-type crystals of this alkane. A chain folding of flat-lying C₃₉₀H₇₈₂ chains, which takes place in few surface locations, was reported only once. Adsorbate of C₂₄₂H₄₈₆ on HOPG is characterized by smooth areas covering the substrate terraces and by few raised drop

lets of the alkane that were not spread on the substrate, Figure 4a. A weak stripped pattern is noticed on smooth areas, and it is more pronounced in the height image of a smaller location, Figure 4b. The white lines of around 5 nm in width and ~0.2 nm in height most likely represent the lamellar edges formed by terminal $-CH_3$ groups. The length of $C_{242}H_{486}$ chains, and this supports the assignment. This image was recorded at low tip-sample forces. At higher forces, the stripped pattern with depressed lines was observed, Figure 4c. This contrast change can be explained by tip-induced depressed of the lamellar boundaries formed by $-CH_3$ terminal groups. In addition, a few of short bright rods are observed on the lamellar surface. They can be assigned to individual alkane chains, which were not incorporated into the lamellar layer. Fine lamellar features were detected at the location, where the lamellar order was perturbed, Figure 4d. In addition to the shift of white lines of the lamellar borders, at some locations inside individual lamellae, there are weak features with smaller spacings. These features are indicated with black arrows, and their separations are ~6 nm. Therefore, a lateral folding of $C_{242}H_{486}$ chains looks possible.

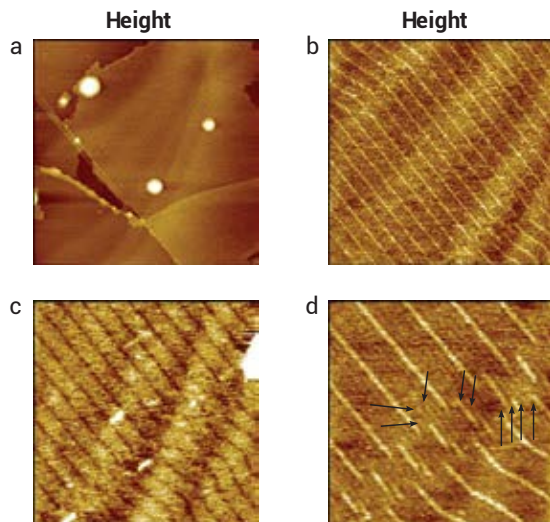


Figure 4.: Height images of $C_{242}H_{486}$ layer on HOPG. The images in (a), (b) and (d) were obtained at low force, and the image in (c) – at higher tip-force. Arrows in (d) point out the fine sub-lamellar features most likely caused by lateral chain folding.

- (a) Image size: $1.5 \times 1.5 \mu m^2$, height range: 11 nm.
- (b) Image size: $0.5 \times 0.5 \mu m^2$, height range: 0.25 nm.
- (c) Image size: $0.4 \times 0.4 \mu m^2$, height range: 0.3 nm.
- (d) Image size: $0.2 \times 0.2 \mu m^2$, height range: 0.25 nm.

Imaging of normal alkanes: $C_{390}H_{782}$

Surface morphology of $C_{390}H_{782}$ layer on HOPG is shown in Figure 5a. Many dark and bright blocks with dimensions of their sides close to 49.5 nm – length of $C_{390}H_{782}$ alkane are detected on this surface. This hints on lamellar order of the top layers. The height image of smaller area (Figures 5b) shows bright features, which can be assigned to the boundaries of lamellae of various size and orientation. The top layer is not complete, and multiple dark vacancies with lamellar dimensions are spread on the surface.

In the attempts to visualize fine details of the lamellar order we have performed imaging at raised tip-forces. The height and phase images, which were recorded at these conditions, revealed depressed lamellar borders as seen in Figures 5c-d. The distance between the bright and dark lamellar edges are close to 49 nm that suggests only small chain tilt with respect to the boundaries.

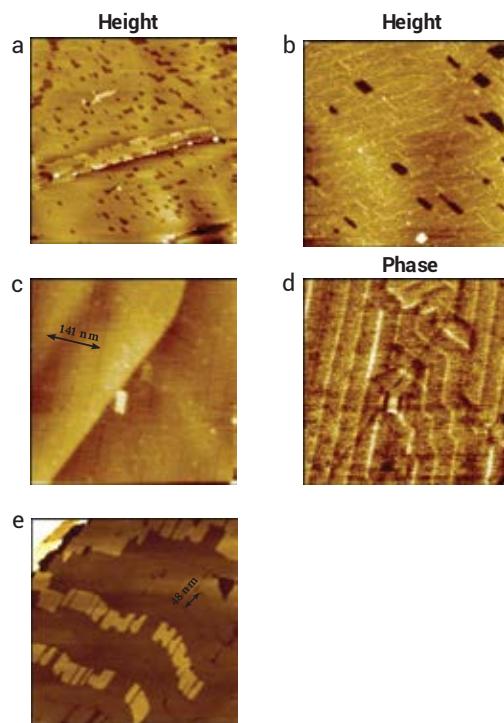


Figure 5.: Height (a,b,c,e) and phase (d) images of $C_{390}H_{782}$ adsorbate on HOPG. The images in (a)-(d) were recorded in dynamic mode in low-force (a-b) and high-force (c-d) operations. The image in (e) was obtained in WaveMode. The inserts in (c) and (e) indicate the width of three and one lamellae, respectively.

- (a) Image size: $2 \times 2 \mu m^2$, height range: 2 nm.
- (b) Image size: $0.6 \times 0.6 \mu m^2$, height range: 0.7 nm.
- (c,d) Image size: $0.5 \times 0.5 \mu m^2$, height range: 0.7 nm, phase range: phase range 5° .
- (e) Image size: $0.5 \times 0.5 \mu m^2$, height range: 4.5 nm.

Fine details in the curved lamellar ribbons were detected in the height image in Figure 5e, which was recorded in WaveMode. This off-resonance mode is characterized by a stronger tip-forces compared to Dynamic Mode. Therefore, more structural features can be visualized. The boundaries of the curved lamellar ribbons are seen with a darker contrast, and individual ribbons exhibit multiple lines with spacings down to 10 nm, which are perpendicular to the boundaries. The parts of the top ribbons epitaxially grown on the underlying lamellae show that they can be split into individual blocks of extended chains. A few dark vacancies with similar dimensions are noticed in the surface as well.

CONCLUSION

In summary, AFM studies of normal and ultra-long alkanes reveal a set of fine structural features related to the lateral chain folding and lamellar architecture in their layers on HOPG. The formation of these nanoscale structures on atomically flat surfaces can be essential for better understanding of alkane and polymer crystallization.

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DriveAFM

Performance without compromise